6.3 The Fundamental Theorem of Calculus

Name _____ Period ____ Date _____

AP Calculus 6.3 (30-minute plan) · Student sheet

I can…

- evaluate a definite integral with an antiderivative.
- find the derivative of a function defined by an integral.
- interpret the integral of a rate of change.

Vocabulary · fill in the missing word

Fundamental Theorem, Part 1 — if f is _____, $\frac{d}{dx}$ ∫ from a to x of f(t) dt = f(x)

Fundamental Theorem, Part 2 — ∫ from a to b of f(x) dx = F(b) − F(a), where F is an _____ of f

Accumulation function — a function like g(x) = _____

Part 1 · Notes and examples (fill in each note, then work the example)

Part 2: if F is an antiderivative of f, then ∫ from a to b of f(x) dx = _____.

Example 1

Evaluate $\int_1^3 (3x^2 - 2x)\,dx$.

Part 1: d/dx ∫ from a to x of f(t) dt = _____.

Example 2

F(x) = ∫ from 2 to x of $\sqrt{t^2 + 9}$ dt. Find F′(4).

With the Chain Rule: d/dx ∫ from a to g(x) of f(t) dt = f(g(x)) · _____.

Net change: the integral of a rate of change gives the net _____ in the quantity.

Example 3

F(x) = ∫ from 1 to x^2 of $(t^3 + 1)$ dt. Find F′(1).

Part 2 · Partner practice

Partner practice

(AP® Practice) If $F(x) = \int$ from 0 to x of sin t dt, then $F'(\frac{\pi}{2}) =$

(A) 1 (B) 0 (C) −1 (D) $\frac{\pi}{2}$